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ORIGINAL RESEARCH

A semi-automated approach facilitated the assessment of the certainty of evidence in a network meta-analysis: Part 1 − Direct comparisons

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Abstract

Objectives: To implement and evaluate a semi-automated approach to facilitate rating the Grading, Recommendation, Assessment, Development and Evaluation (GRADE) certainty of evidence (CoE) for direct comparisons within two living network meta-analysis.

Methods: For each of three GRADE domains (study limitations, indirectness, and inconsistency), decision rules were developed and used to generate automated judgments for each domain and the overall certainty. Inputs included risk of bias and indirectness ratings for each study and measures of heterogeneity. Indirectness ratings were made by two independent reviewers and resolved through consensus. With the help of an online tool (customized to our project), two independent raters viewed forest plots and additional data and could confirm or modify the suggested rating. Disagreements were resolved by consensus. We evaluated inter-rater reliability and accuracy.

Results: Across 374 direct comparisons, there was perfect agreement (100%) between the automated judgment and reviewer consensus, when only a single study was available ($n = 292$), and near-perfect agreement when more than one study was available (99%−100% for the three GRADE domains and 96% for overall rating). Inter-rater reliability was near perfect (Gwet's AC1 kappa score ranging from 96% to 100%).

Conclusion: Automated judgments using established decision rules agreed with expert judgment for the vast majority of GRADE CoE ratings. © 2025 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY-NC license (http://creativecommons.org/licenses/by-nc/4.0/).

Keywords: Randomized controlled trials (RCTs); Grading of recommendation, assessment; development and evaluation (GRADE); Network meta-analysis (NMA); Rheumatoid arthritis (RA); Systematic reviews; Living systematic reviews; Semi-automation

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What is new?

Key findings

- Most GRADE domain and overall ratings for direct evidence can be auto-generated accurately.

What this adds to what is known?

- This study shows benefits of a semi-automated approach to assess direct evidence certainty.
- The approach includes a study-level evaluation of indirectness.

What is the implication and what should change now?

- Evaluating the CoE can use semi-automation to facilitate expert judgements.
- The online tools for indirectness and domain judgements are viewable but project specific.

1. Background

The Grading of Recommendation, Assessment, Development, and Evaluation (GRADE) is a widely used framework for assessing the certainty of evidence (CoE) and strength of recommendations in health care [1]. It provides a systematic and transparent approach to evaluating CoE, considering the domains of risk of bias (ROB), inconsistency, indirectness, imprecision, and publication bias [1−3]. GRADE uses a structured process to rate the CoE across a body of evidence. It classifies evidence into four levels: high, moderate, low or very low, based on the assessment of these key domains [2,4].

Network meta-analysis (NMA) are increasing in popularity, as they allow comparisons between all available treatments. Direct evidence (where treatments have been directly compared) and indirect evidence (where treatments can be compared indirectly through a common comparator) are both included to generate overall estimates of effect between every pair of interventions [5,6]. In the GRADE approach for NMAs, the CoE is first assessed for the direct and indirect evidence separately [7]. A certainty rating for the overall NMA estimate is then made based on the ratings of the direct and indirect evidence, including an assessment of whether there are concerns about an imbalance of effect modifiers between the direct evidence that inform the indirect evidence (intransitivity), an important difference in direct and indirect effect estimates (incoherence), and consideration of imprecision of the final NMA estimate [7,8].

One of the challenges associated with assessing the CoE in an NMA is the workload required [8,9]. A review with 10 treatments and seven outcomes will have 315 (45 × 7) pairwise comparisons, and each of these may have direct and/or indirect evidence that each need to be assessed, in addition to the overall NMA estimate. Living systematic reviews (LSR) and NMAs present an additional challenge, as the certainty ratings need to be updated when a new trial is included, and review teams may change over time [10−12]. Processes that provide explicit rationale for the required judgments may improve transparency and facilitate consistency in the judgments over time [13].

Our objective was to implement and evaluate a semi-automated approach to facilitate rating the GRADE CoE for direct comparisons within two living NMAs for rheumatoid arthritis (RA) drug therapy. In this paper, we focused on steps related to assessing the certainty of the direct evidence; later work will explore similar efforts for the indirect evidence and overall NMA estimates. Importantly, our approach recognizes the inherent subjectivity of decisions as to whether to rate the CoE down, but also that these decisions should be informed, as much as possible, by explicit criteria. We use the term ‘‘semi-automated’’ as any automatically generated judgment that still required confirmation by reviewers.

2. Methods

2.1. Overview of approach

An overview of our approach and how each step integrates with the overall GRADE-NMA process are presented in Figure 1. Our approach used predefined criteria to generate suggested CoE ratings for the GRADE domains, ROB, inconsistency, and indirectness [14]. Reviewers could confirm or modify these ratings using a custom online tool we built that displayed the suggested rating and relevant data. The predefined criteria were developed through expert consensus following GRADE guidance. They were developed with our specific review in mind rather than as universal rules, as we recognized that the criteria used might differ by review. Also to facilitate our approach, we included a rating of ‘‘some concerns’’ for the various domains to provide further nuance for GRADE assessment. We did not consider the domains of publication bias and imprecision for this study. Publication bias is often assessed externally, for example, by evaluating whether all registered trials are published, and we had a few studies in each comparison to explore the presence of small study effects [15]. While imprecision judgments could be easily automated, GRADE guidance suggests that the assessment of imprecision in an NMA can often be skipped for the direct estimate and evaluated at the NMA or mixed evidence phase [7].

2.2. Data sources and analyses

We used data from two Cochrane LSRs and NMAs of drug therapy in RA. The reviews have not yet been

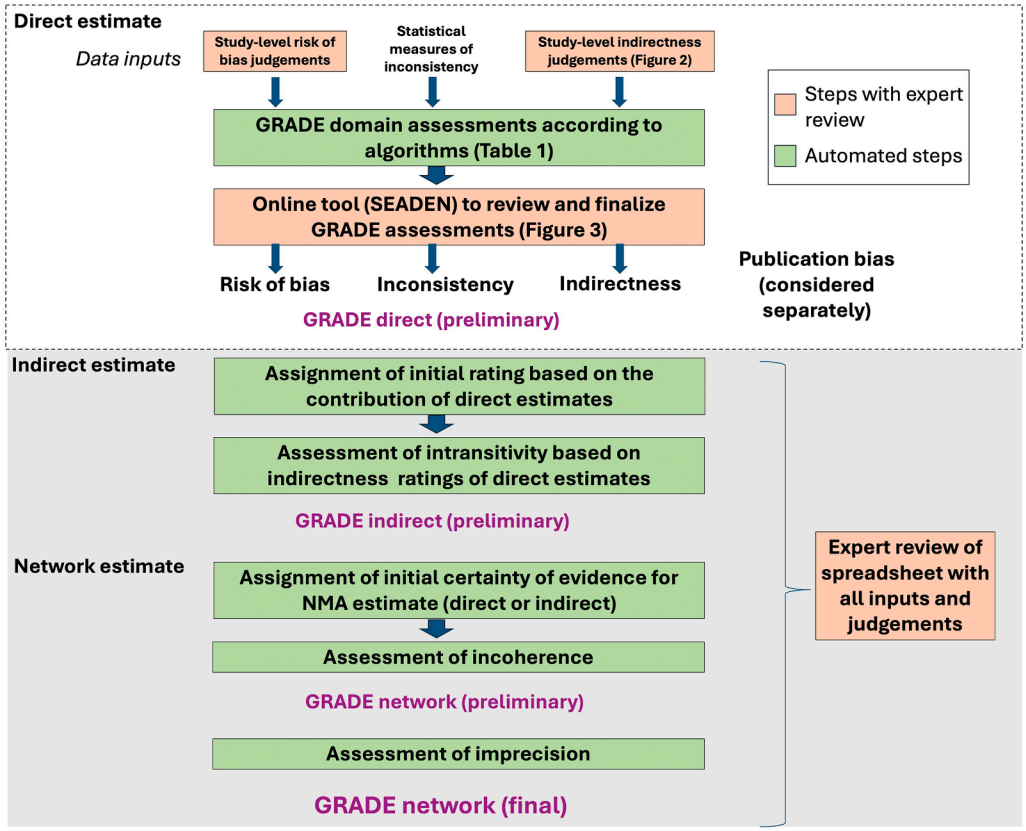

Figure 1. Overview of our approach within the overall GRADE-NMA process. This paper focuses on generating the certainty rating for the direct estimate.

published in full, but the GRADE evidence table for NMA#1 (defined below) has been published and used to inform living guidelines in Canada and Australia [16,17]. The methods are fully described in our published protocol [18]. Although the main analysis in the review excludes studies at high ROB (and includes them in a sensitivity analysis), for this paper, studies flagged as "high" ROB were included to add nuance and broaden the scope of the evaluated direct evidence. The NMAs compare different disease-modifying anti-rheumatic drug (DMARD) options as initial therapy (NMA #1) or after an inadequate response to anti-tumor necrosis factor (TNF) therapy (NMA #2). Both reviews consider the same seven outcomes, (additional details of the outcomes and the analyses specifications are provided in Appendix 1 and the published protocol [18]).

2.3. Data inputs

2.3.1. Risk of bias

As outlined in the protocol [18], each trial had received an overall ROB rating of "low," "unclear," or "high" for each of three different outcome types (radiographic progression, withdrawals, and all other clinical outcomes) using the original 2011 Cochrane ROB tool [19]. These ratings had been made independently by two reviewers and consensus with an expert if needed, following a detailed review-specific guide we developed (Appendix 2a-Table 1). An overall rating of "unclear" meant there were reasons to be concerned about biases in the study but not sufficient to warrant a "high" rating. In using the ROB judgments for the automated GRADE ratings, we used the term "some concerns" instead of "unclear" to be consistent with terminology used in the updated Cochrane ROB-II tool and GRADE [20].

2.3.2. Indirectness

For the indirectness domain, we applied an approach adapted from the Confidence in Network Meta-Analysis (CINeMA) framework to assign an indirectness rating to individual trials [21,22]. We used the same ratings of "high," "some concerns," and "low" as with ROB. Following GRADE guidance [23,24], the indirectness ratings were made considering the characteristics of the population, intervention (including co-interventions), or comparator(s) in the trial that could plausibly modify the effect of the intervention relative to usual modern clinical practice [24]. We focused on key trial characteristics that had either been previously found to be an effect-modifier in our prior NMA (Appendix 2a-Table 2) [25] or were thought to have potentially important effect modification, based on the expert opinion of clinicians in our team. Ratings were assigned independently by two reviewers using a custom

Table 1. Rules for automated certainty of evidence (CoE) judgments

Criterion		Domains				
		Risk of bias (ROB) and indirectness			Inconsistency	
Rule		Weight of studies with "high" ratings >50%	Weight of studies with "high" ratings 30% to 50%	Weight of studies with "some concerns" ratings >50%	I^2 > 50%	I^2 between 30% and 50%
Rating	ROB	Very serious (↓ two levels)	Serious (↓ one level)	Some concerns	Serious (↓ one level)	Some concerns
	Indirectness	Serious (↓ one level)	Some concerns	Some concerns		

online platform developed using the Shiny package in R software [26], where reviewers were shown the relevant data from the trial and assigned their judgment (see Fig 2 for a screenshot). All ratings are of "high" and "some concerns," and all disagreements were discussed as a team for consensus.

2.3.3. Inconsistency

The main inputs for the inconsistency or heterogeneity ratings were the Higgins' I^2 values. Additionally, we also calculated and displayed τ^2 values and forest plots for raters to review when confirming the judgments. We estimated the I^2 based on the heterogeneity Q-statistic [27] and Tau^2 using the Paule-Mandel estimators using the *meta* package in R that imports the *metafor* package referenced in the Cochrane Handbook (Chapter 10) [28–30]. We did not assess prediction intervals, which are typically only informative when the meta-analysis includes five or more studies [31].

2.4. Automated ratings

2.4.1. Domain ratings

Using the data inputs described above, we developed rules for assigning automated judgments for each domain by consensus of our team (Table 1). For the ROB and indirectness domains, judgments were assigned based on the individual study ratings and their percent contribution [28] to the pooled estimate. For the ROB domain, we rated down one level ("serious"), if the percentage contribution from

Table 2. Summary of direct comparisons available in each NMA

Review	Outcome	Outcome type	Number of direct comparisons				
			One study	Two studies	Three studies	More than three studies	Total
DMARD-naive	ACR50	Dichotomous	40	8	3	1	52
	DAS28	Continuous	25	-	2	1	28
	Pain	Continuous	26	5	-	2	33
	HAQ-DI	Continuous	38	7	5	2	51
	Radiographic progression (change from baseline)	Continuous	28	8	1	-	37
	Radiographic progression (final value)	Continuous	6	1	-	-	7
	Serious adverse events	Rate	28	6	3	1	38
	Withdrawal due to adverse events	Rate	32	11	5	3	51
TNF-IR	ACR50	Dichotomous	16	2	-	-	18
	DAS28	Continuous	7	-	-	-	7
	Pain	Continuous	5	-	-	-	5
	HAQ-DI	Continuous	17	2	-	-	19
	Radiographic progression	Continuous	-	-	-	-	-
	Serious adverse events	Rate	12	2	-	-	14
	Withdrawal due to adverse events	Rate	12	2	-	-	14

ACR50, American College of Rheumatology response criteria-50, a composite measure of disease improvement; DAS28, Disease Activity Score in 28 joints, a continuous measure of disease activity; HAQ-DI, Health Assessment Questionnaire-Disability Index, a continuous measure of functional disability; DMARD-naïve, disease-modifying anti-rheumatic drug; TNF-IR, tumor necrosis factor—inadequate responder.

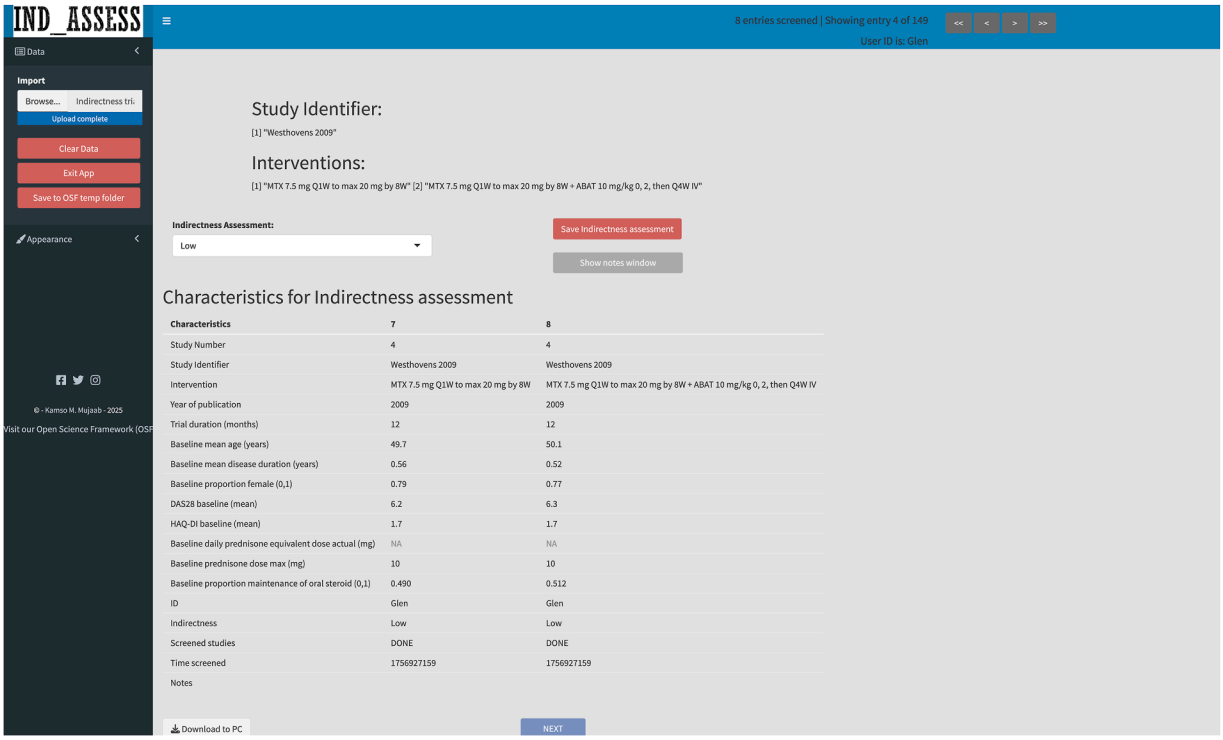

Figure 2. Screenshot of the study-specific indirectness application.

studies at high or some concerns ROB was 30%-50%, and two levels (“very serious”), if the percentage contribution was greater than 50%. These thresholds were informed by GRADE guidance, where rating down by two levels suggests that most studies should have significant limitations [14]. We assigned an overall rating of “some concerns” for the ROB domain if neither of these criteria were met, but the cumulative percentage contribution of trials with ratings of “some concerns” was >50%. For the indirectness domain, we used similar thresholds but were more conservative in rating down (Table 1), as GRADE guidance advises caution when considering a downgrade for indirectness [23,24,32]. We assigned ratings of “some concerns” and “serious” if the percentage contribution from studies with a high indirectness rating was 30%-50% and >50%, respectively (Table 1).

Finally, for the inconsistency domain, the automated certainty rating was “serious” (downgraded one level), if the Higgins I^2 value was greater than 50%, and “some concerns,” if the I^2 value fell within the range of 30%-50%. We chose these thresholds, based on Cochrane guidance, where >50% indicates substantial heterogeneity and 30% represents the lower threshold for moderate heterogeneity [33,34].

2.4.2. *Overall certainty rating*

The final rating assignment involved an algorithmic accumulation of the ratings across the three domains (ROB, indirectness, and inconsistency). Numeric values were assigned to each rating (“not serious” = 0, “some concerns” = 0.5, “serious” = 1, “very serious” = 2) and then summed and rounded down to determine how many times to downgrade the evidence, broadly aligning with the GRADE guidance [3]. We added an additional level of “extremely low” to the final certainty rating, to document situations where the evidence was downgraded more than three times and provide transparency to reviewers who could then reflect on whether this was appropriate. In presenting the results in our final review, this “extremely low” category is collapsed with “very low,” to align with the GRADE guidance.

2.5. *Manual review of automated judgments*

Every automated judgment was reviewed independently by two reviewers (GH, SW), both of whom are rheumatologists with extensive experience in rating the CoE using GRADE. Following the individual reviewer judgments, the research team discussed disagreements between reviewers and came to a consensus.

To facilitate the individual reviewer judgments, we developed an online tool using the Shiny package in R software [26]: semi-automated assessment of direct evidence from a network meta-analysis (SEADEN) application (screenshot in Fig 3). The tool displayed a forest plot for each direct comparison and asked reviewers to confirm or modify the suggested certainty rating (screenshot in Fig 3). The forest plot included the ratings for ROB, indirectness, the percentage weight of each study to the pooled estimate, and

measures of heterogeneity (Higgins I^2 and τ^2). Below the forest plot, automated judgments for each of the three domains were shown in a drop-down box that reviewers could change to a different rating, as well as provide free-text comments in the box to justify their decision. Details of the reviewer's process for domain ratings and certainty judgments are provided in Appendix 2b, and data are hosted to an Open Science Framework (OSF) website [35].

2.6. Evaluation of approach

As the indirectness ratings were a new step in our approach (as opposed to ROB ratings that are standard in reviews), we calculated inter-rater reliability. This was done using percent agreement and Gwet's Aggrement Coefficient (AC) 1 kappa scores, which are a more robust measure for ordinal ratings [36]. The interpretation of kappa values followed established guidelines: no agreement ($\leq$0), slight or fair agreement (0.01−0.40), moderate agreement (0.41−0.60), substantial agreement (0.61−0.80), and near-perfect or perfect agreement (0.81−1.00) [16,36,37].

We evaluated the agreement between the automated and final expert ratings by calculating the percentage of times that the reviewer judgments agreed with the automated ratings for each domain and the overall certainty rating. This was done for both the individual reviewer judgments and the consensus judgment, according to the number of studies present for the direct comparison. The inter-rater reliability

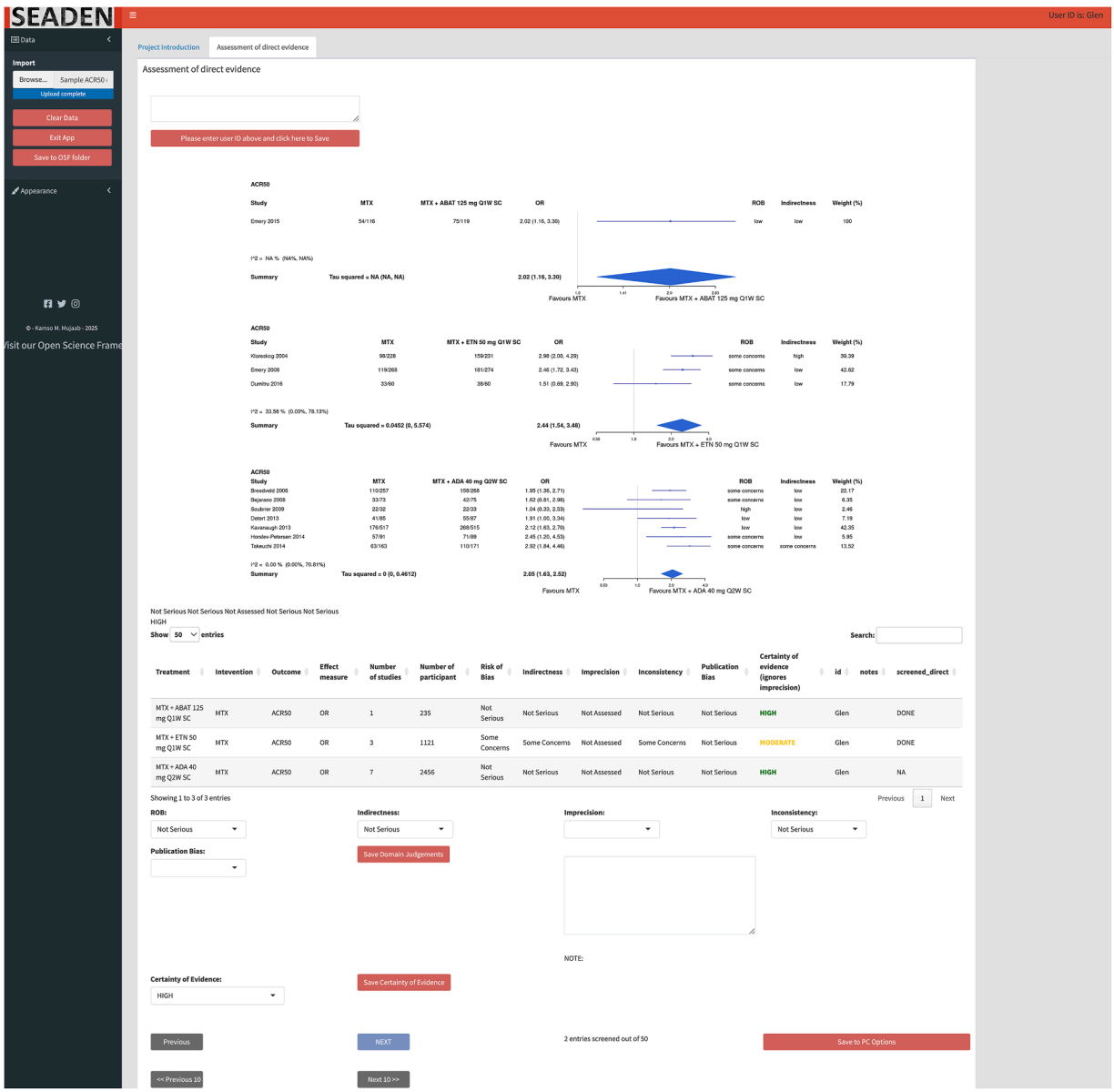

Figure 3. Screenshot of the semi-automated assessment of direct evidence from a network meta-analysis (SEADEN) application from R-shiny dashboard.

for the domain and overall judgments was made using the same metrics described above. All analyses were conducted in R software.

3. Results

3.1. Indirectness ratings

Of the 112 studies evaluated for indirectness, 66 (58.9%) were rated as "low," 21 (18.8%) as "some concerns," and 25 (22.3%) as "high," with a percentage agreement of 80.4% and Gwet's AC1 kappa score of 72.9% (62.0%, 83.7%), indicating substantial agreement between the two raters. Indirectness concerns largely involved older trials, where the treatment of RA was very different from contemporary RA management. This included trials where patients had an extremely long disease duration for a DMARD-naïve population (eg, >5 years), were treated as an inpatient for their RA, or were using very low doses of methotrexate. Through group discussion, we developed two consensus rules based on disease duration and methotrexate dose (a key co-intervention), which we incorporated in our detailed guidance document for the living NMAs (Appendix 2a).

3.2. GRADE certainty ratings

3.2.1. Characteristics of included comparisons

Out of 374 direct comparisons across all outcomes from both reviews, 292 (78%) were from a single study, 54 (14.4%) included two studies, 19 (5.1%) included three studies, and 10 (2.7%) included more than three studies (Table 2). Of these direct comparisons, 152 (41%) were for continuous outcomes, 117 (31%) for rate outcomes, and 105 (28%) for dichotomous outcomes. With 374 direct comparisons, there were 1496 individual judgments (374 × 4) across the three domains and overall rating.

3.2.2. Analyses of approach

Across all 1496 judgments, the consensus judgment agreed with the automated rating of 1486 times (99.2%), with a Gwet's AC1 kappa score of 99.1% (98.0%, 100%), indicating near-perfect agreement (Table 3). Between reviewers, there was 98.7% agreement and a Gwet's AC1 kappa score of 98.5% (97.1%, 99.8%). When only a single study was present, the automated judgment had perfect agreement with both reviewers across all domains. When more than one study was present, agreement between the consensus judgment and automated rating was also very high at 96.4% for the overall certainty rating and between 98.8% and 100% for the three GRADE domains (Appendix 3-Table 3). Similar levels of agreement were seen between reviewers (Appendix 3-Table 3).

Details on the 10 instances of disagreement between reviewers or where the consensus judgment differed from the automated rating are outlined in Appendix 4. Disagreements occurred once in the ROB domain, once in the inconsistency domain, and three times in the indirectness domain, leading to differences in the overall rating on five occasions. These discrepancies often stemmed from one reviewer adopting a more cautious approach, emphasizing the similarity of the pooled effect's point estimate to that of the most influential study, suggesting that the pooled estimate should reflect the rating of this study.

4. Discussion

In this study, we found that 96% of domain and overall GRADE certainty ratings for direct evidence can be automatically generated using clear decision rules. Our approach requires individual study-level ratings for indirectness, which is typically not part of systematic reviews but is a useful addition at the data extraction phase. We developed web tools to implement our approach, displaying necessary information with suggested ratings, while allowing reviewers to override and document their rationale. This flexibility ensures that nuanced judgments are retained, which can be used to develop new rules in the algorithm. While our approach could be applied to any traditional pairwise meta-analysis, it may be particularly useful in NMAs, where a large number of judgments need to be made. This is especially relevant for living NMAs, where clear decision rules can facilitate consistent ratings over time while reducing workload. Moreover, these assessments serve as

Table 3. Reliability assessment for the direct evidence judgments

Comparison	Statistic	Domains			Overall certainty rating[a]
		Risk of bias (ROB)	Indirectness	Inconsistency	
Between reviewers	Gwet's AC1 inter-rater reliability (CI)	99.3% (98.4%, 100.0%)	98.7% (97.3%, 100.0%)	99.4% (98.6%, 100.0%)	98.5% (97.1%, 99.8%)
	Percentage agreement	99.5%	98.9%	99.5%	98.7%
Consensus judgment and automated ratings	Gwet's AC1 inter-rater reliability (CI)	99.7% (99.0%, 100.0%)	100.0% (NA)	99.7% (99.2%, 100.0%)	99.1% (98.0%, 100%)
	Percentage agreement	99.7%	100.0%	99.7%	99.2%

CI, confidence interval; NA, not applicable.
[a] Overall rating considering the three domains (excluding publication bias and imprecision).

the building blocks for evaluating indirect and network evidence, ensuring a cohesive and streamlined process.

Our approach does not replace expert judgment, which is essential for CoE assessments [38]. Expert judgment is incorporated at two stages: 1) developing rules and algorithms and 2) reviewing automatically assigned judgments. Our web tool (SEADEN) ensured expert review of every judgment, although opportunities may exist to reduce reviewer burden. Particularly, when only one study was present, no reviewer overrode any automatic judgments, as there was no need to balance ratings. However, when multiple studies are present, reviewers need to consider more nuances. For example, if high and low ROB studies show similar results, it may be appropriate to not downgrade, even if the high ROB studies contribute more than 30% (our chosen threshold) to the pooled estimate. Over time, algorithm tweaks may capture nuances identified in consensus discussions. However, fully automating rating is not the goal, as expert confirmation ensures the credibility of certainty ratings.

Our approach leverages the GRADE framework to assess CoE in NMA, incorporating study-specific indirectness ratings from the CINeMA tool [21,22]. While CINeMA typically assigns these ratings to the overall NMA estimate, we found them useful for assigning CoE judgments to direct evidence, aiding automation. Decisions to rate down for indirectness are review-specific, but establishing rules and implementing these aided in the consistency. Our work contributes to automation in systematic reviews, although most existing research is focused on trial identification and data extraction [39−42]. Recent GRADE publications propose automation strategies to address NMA workload, including a spreadsheet-based CoE rating approach [9]. Our study introduces a novel method for semi-automating judgments for direct evidence, aligning with a ‘‘human-in-the-loop’’ approach, where humans make the final judgment [43].

Strengths of our study include high accuracy in semi-automated judgments across two NMAs, supporting its continued use. Establishing defined rules and having reviewers document disagreements may improve transparency and consistency, especially in LSR where teams may change [44]. Our approach is applicable across networks of all types and sizes, as demonstrated by the range of evaluated outcomes. However, a limitation is that the online tools developed are customized to our project. A living NMA provided the justification for this work, as the upfront time to develop them will remain valuable in our efforts over time. However, additional work would be required to make them generalizable for broader use. Even without these tools, the principles of our approach can still be applied. We did not address publication bias or imprecision, so the final ratings cannot be used alone. Finally, despite our ‘‘human-in-the-loop’’ approach, automating judgments may reduce thoughtful consideration, so reviewers should have GRADE expertise to ensure accurate, consistent judgments. The extraction phase remains resource-intensive, and our approach may further increase workload by including study-level indirectness judgments that require careful consideration of effect modifiers. The reliability assessment may be biased because reviewers knew that their agreement with automated assessments was being measured, potentially inflating estimates; blinded evaluations would be more robust.

Future research should explore the usefulness and accuracy of our tools and approach to other reviews and in blinded comparison to traditional GRADE ratings. Additionally, we believe that there would be value in developing a formal tool to assess the indirectness of individual studies. While the specific aspects of the population, intervention, comparison, and outcome (PICO) elements considered would be review-specific, a generic validated tool could aid in assigning the judgments. Finally, making our custom online tools universally applicable for any review would be a useful next step.

5. Conclusion

Automation and established decision rules can be used to implement most judgments when evaluating the CoE from an NMA. In addition to aiding in the efficiency of the process, this may also help maintain transparency and consistency of judgments over time.

CRediT authorship contribution statement

Mohammed Mujaab Kamso: Writing − review & editing, Writing − original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Samuel L. Whittle:** Writing − review & editing, Resources, Conceptualization. **Jordi Pardo Pardo:** Writing − review & editing, Investigation, Conceptualization. **Rachelle Buchbinder:** Writing − review & editing, Resources, Funding acquisition, Conceptualization. **George Wells:** Writing − review & editing, Resources, Funding acquisition, Conceptualization. **Rob Deardon:** Writing − review & editing, Supervision, Resources, Conceptualization. **Tolulope Sajobi:** Writing − review & editing, Methodology, Conceptualization. **George Tomlinson:** Writing − review & editing, Methodology, Conceptualization. **Jesse Elliott:** Writing − review & editing, Resources, Conceptualization. **Jocelyn Thomas:** Writing − review & editing, Resources, Conceptualization. **Shannon E. Kelly:** Writing − review & editing, Resources, Conceptualization. **Romina Brignardello-Petersen:** Writing − review & editing, Conceptualization. **Glen S. Hazlewood:** Writing − review & editing, Writing − original draft, Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jclinepi.2025.112109.

Data availability

Data will be made available on request.

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